

Question number	Answer	Additional guidance	Mark
6(a)	An answer that includes the following points: • number of (different) species (1) • genetic diversity within a species (1)	ACCEPT species richness amount ACCEPT variation in {genotypes / alleles}	(2)

Question number	Answer	Additional guidance	Mark
6(b)	An explanation that includes two of the following points: • <u>absorb</u> light energy so that electrons are {excited / released} (1) • to synthesise ATP and reduced NADP (1)	ACCEPT NADPH IGNORE ⁺ signs ALLOW to absorb light energy so that it can be converted into ATP energy = 1 mark	(2)

Question number	Answer	
(c)(i)	Indicative content: Comparisons: - propanone extracts more chlorophyll a from species P - propanone extracts more chlorophyll a from species R - DMSO extracts more chlorophyll a from species Q than propanone - propanone and DMSO extract similar concentrations of chlorophyll b from species S - propanone extracts more chlorophyll b from species P - propanone extracts more chlorophyll b from species Q - propanone extracts more total chlorophyll from species P Generalisations: - propanone is the most effective solvent at extracting chlorophyll - species P appears to contain the most chlorophyll when using propanone - DMSO is generally less effective than propanone except when extracting chlorophyll a from species Q Implications in identifying species: - some chlorophyll lost when extracted together as total is less than the sum of the components - other pigments extracted in some cases as total is more than individual components added together - the choice of solvent depends on the chlorophyll / species being extracted - because of difference in solubility (of pigments / membranes) / permeability of membranes - more than one solvent needs to be used if this method is to be used for identifying species - as different solvents extract different concentrations of different chlorophylls from different species - possibility of looking at extraction of other pigments - some sort of comparison table / calibration curve needed to match profile to extraction profiles - propanone better if only using one solvent as results the most varied - avoids need for DNA analysis - comparisons can be made in the field / with simple equipment - no indication of validity of data	Level 1 : 1 mark = 2 comparisons listed 2 marks = 4 comparisons listed Level 2 : 3 marks = 6 comparisons OR 1 generalisation + 3 comparisons OR 2 generalisations 4 marks = one implication discussed Level 3 : 5 marks = two implications discussed 6 marks = three implications discussed

Question number	Answer	Additional guidance	Mark
6(c)(ii)	An answer that includes two of the following points: • chlorophylls have different solubility in different solvents (1) • because the chlorophylls have different structures (1) • different solvents can {permeate / dissolve / disrupt} (cell / chloroplast) membranes differently (1)	ACCEPT more / less will dissolve IGNORE Rf values reacted with solvent	(2)